Pairwise Equal Charge Solution in Kerr-Schild Form

After failing to express the 2-charge pairwise equal charge solution in the ansatz structure from Wu, we attempt to express the pairwise equal charge solution in a Kerr-Schild-like form. That is, we attempt to express the metric in the form

$$ds^2 = d\bar{s}^2 + \frac{2m}{U}(k_\mu \, dx^\mu)^2 \tag{1}$$

where $d\bar{s}^2$ is a AdS$_4$ background metric we expect from the massless limit and the 1-form k_μ is linear to the electric gauge potentials.

The logic we follow in this section is quite straight forward – alike how Wu has sought the Cvetic-Youm solutions, I look at the $g = 0$ limit to obtain the appropriate coordinate transformation.

Proposition. Upon the following reparametrisation of the ungauged metric ($g = 0$) with charge

$$R^2 = (r + 2ms_1^2)(r + 2ms_2^2) \tag{2}$$

we arrive at the following form (the new coordinate R is simply written as r)

$$\begin{aligned} ds^2 =& - dt^2 + (r^2 + a^2\cos^2\theta)\Big(\frac{dr^2}{\Delta_r} + d\theta^2\Big) + (a^2 + r^2)\sin^2\theta \, d\phi^2 \\ &+ \frac{2mU}{r^2 + a^2\cos^2\theta}(dt - a\sin^2\theta \, d\phi)^2 \end{aligned} \tag{3}$$

with

$$\begin{aligned} \Delta_r =& (r^2 + m^2(s_1^2 - s_2^2)^2)\Big[a^2 + r^2 - 2mU\Big]/r^2 \\ U =& (1 + s_1^2 + s_2^2)\sqrt{r^2 + m^2(s_1^2 - s_2^2)^2} - m[s_1^2(1 + s_1^2) + s_2^2(1 + s_2^2)] \end{aligned} \tag{4}$$

The metric shows similaries with that of the ungauged, uncharged Wu form:

$$\begin{aligned} ds^2 =& - dt^2 + (r^2 + a^2\cos^2\theta)\Big(\frac{dr^2}{r^2 + a^2 - 2mr} + d\theta^2\Big) + (a^2 + r^2)\sin^2\theta \, d\phi^2 \\ &+ \frac{2mr}{r^2 + a^2\cos^2\theta}(dt - a\sin^2\theta \, d\phi)^2 \end{aligned} \tag{5}$$

Proposition. In the case of the gauged case ($g \neq 0$), the result is similar. Upon the same reparametrisation, we arrive at

$$\begin{aligned} ds^2 =& -\frac{(1+g^2r^2)(1-a^2g^2\cos^2\theta)}{1-a^2g^2}\,dt^2 \\ &+ (r^2+a^2\cos^2\theta)\Big(\frac{dr^2}{\Delta_r}+\frac{d\theta^2}{1-a^2g^2\cos^2\theta}\Big) \\ &+ \frac{r^2+a^2}{1-a^2g^2}\sin^2\theta\,d\phi^2 + \frac{2mU}{r^2+a^2\cos^2\theta}\Big(\frac{1-a^2g^2\cos^2\theta}{1-a^2g^2}\,dt - \frac{a}{1-a^2g^2}\sin^2\theta\,d\phi\Big)^2 \end{aligned} \tag{6}$$

with

$$\Delta_r = (r^2+m^2(s_1^2-s_2^2)^2)\Big[(r^2+a^2)(1+g^2r^2)-2mU\Big]/r^2 \tag{7}$$

and the same U defined above. This again shows a resemblance to the gauged but uncharged solution:

$$\begin{aligned} ds^2 =& -\frac{(1+g^2r^2)(1-a^2g^2\cos^2\theta)}{1-a^2g^2}\,dt^2 \\ &+ (r^2+a^2\cos^2\theta)\Big(\frac{dr^2}{(r^2+a^2)(1+g^2r^2)-2mr}+\frac{d\theta^2}{1-a^2g^2\cos^2\theta}\Big) \\ &+ \frac{r^2+a^2}{1-a^2g^2}\sin^2\theta\,d\phi^2 + \frac{2mr}{r^2+a^2\cos^2\theta}\Big(\frac{1-a^2g^2\cos^2\theta}{1-a^2g^2}\,dt - \frac{a}{1-a^2g^2}\sin^2\theta\,d\phi\Big)^2 \end{aligned} \tag{8}$$

Convention Matching

With the two independent charge solution perfectly fitting into the Wu form, we now attempt to fit the four charge ungauged case into the Wu form. For this to be properly set up, we first need to match conventions. Using the duality transformation

$$\begin{cases} F^2 = e^{-\varphi_1+\varphi_2-\varphi_3} \star (\tilde{\mathcal{F}}_2 + \chi_3 e^{2\varphi_3}\mathcal{F}^1) - \chi_1(\tilde{\mathcal{F}}_3 + \chi_3 F^4) \\ F^3 = e^{-\varphi_1-\varphi_2+\varphi_3} \star (\tilde{\mathcal{F}}_3 + \chi_2 e^{2\varphi_2}\mathcal{F}^1) - \chi_1(\tilde{\mathcal{F}}_2 + \chi_2 F^4) \end{cases}$$

Coefficient Determination Algorithm

At the current stage of the project, there is a need for the determining the coefficients of the gauge-linear terms. In this section, I suggest that the method of achieving this is by preforming a Taylor series expansion in terms of the charges. Here, we consider the following examples (all of which were proven to be in the Wu form). Ultimately, we will want to apply this logic to the four charge ungauged solution.

(i) Three charge AdS_5 solution

(ii) Two charge AdS_4 solution

Example. (Two charge AdS_4) We first present the explicit form of the solution and the gauge fields.

$$\begin{aligned} ds^2 = (H_1 H_2)^{1/2} \Big[& - dt^2 + \Sigma \Big(\frac{dr^2}{r^2 + a^2 - 2mr} + d\theta^2 \Big) + (r^2 + a^2) \sin^2\theta \, d\phi^2 \\ & + \frac{s_1^2}{(s_1^2 - s_2^2)} \Big(\frac{1}{1 - 1/H_1} \Big) A_1^2 + \frac{s_2^2}{(s_2^2 - s_1^2)} \Big(\frac{1}{1 - 1/H_2} \Big) A_2^2 \Big] \end{aligned}$$

Meanwhile, the gauge fields are given as

$$A_i = \Big(1 - \frac{1}{H_i} \Big) \Big(\frac{c_i}{s_i} \, dt - \frac{c_1 c_2}{s_i c_i} a \sin^2\theta \, d\phi \Big)$$

where

$$H_i = 1 + \frac{2mr s_i^2}{\Sigma} \quad c_i = \sqrt{1 + s_i^2}$$

Preforming a Taylor expansion of the terms inside the warp factor with respect to s_2 we find (up to $O(s_2^4)$ terms and subtracting the pure AdS terms)

$$\begin{cases} O(0): & \frac{1}{s_1^2} \Big(1 - \frac{1}{H_1} \Big) (c_1 dt - c_2 a \sin^2\theta \, d\phi)^2 \\ O(s_2^2): & \frac{1}{s_1^4} \Big(1 - \frac{1}{H_1} \Big) (c_1 \, dt - c_2 a \sin^2\theta \, d\phi)^2 - \frac{2mr}{\Sigma s_1^2} (c_2 \, dt - c_1 a \sin^2\theta \, d\phi)^2 \end{cases}$$

Doing the same for s_1,

$$\begin{cases} O(0): & \frac{1}{s_2^2} \Big(1 - \frac{1}{H_2} \Big) (c_2 dt - c_1 a \sin^2\theta \, d\phi)^2 \\ O(s_1^2): & \frac{1}{s_2^4} \Big(1 - \frac{1}{H_2} \Big) (c_2 \, dt - c_1 a \sin^2\theta \, d\phi)^2 - \frac{2mr}{\Sigma s_2^2} (c_1 \, dt - c_2 a \sin^2\theta \, d\phi)^2 \end{cases}$$

Example. (Three charge AdS_5 solution) We first present the explicit form of the solution and the gauge fields.

$$\begin{aligned} ds^2 = (H_1 H_2 H_3)^{1/3} \Big[& - dt^2 + \Sigma \Big(\frac{r^2 \, dr^2}{(r^2 + a^2)(r^2 + b^2) - 2mr} + d\theta^2 \Big) + (r^2 + a^2) \sin^2\theta \, d\phi^2 \\ & + (r^2 + b^2) \cos^2\theta \, d\psi^2 + \frac{s_1^2}{(s_1^2 - s_2^2)(s_1^2 - s_3^2)} \Big(\frac{\Sigma H_1}{2m} \Big) A_1^2 \\ & + \frac{s_2^2}{(s_2^2 - s_1^2)(s_2^2 - s_3^2)} \Big(\frac{\Sigma H_2}{2m} \Big) A_2^2 + \frac{s_3}{(s_3^2 - s_1^2)(s_3^2 - s_2^2)} \Big(\frac{\Sigma H_3}{2m} \Big) A_3^2 \Big] \end{aligned}$$

Meanwhile, the guage fields are given as

$$A_i = \frac{2m}{\Sigma H_i} \Big[\frac{s_i c_1 c_2 c_3}{c_i} \Big(\frac{c_i^2}{c_1 c_2 c_3} dt - a \sin^2\theta \, d\phi - b \cos^2\theta \, d\psi \Big) + \frac{c_i s_1 s_2 s_3}{s_i} \Big(b \sin^2\theta \, d\phi + a \cos^2\theta \, d\psi \Big) \Big]$$

where

$$H_i = 1 + \frac{2ms_i^2}{\Sigma} \quad c_i = \sqrt{1+s_i^2}$$

Preforming a Taylor expansion of the terms inside the warp factor with respect to (s_2, s_3) we find (up to $O(s_2^4, s_3^4)$ terms and subtracting the pure AdS terms)

$$O(0): \frac{2m}{\Sigma}\Big[\frac{1}{H_1}\frac{1}{s_1^2}\Big(c_1\,dt - \frac{c_1c_2c_3}{c_1}(a\sin^2\theta\,d\phi + b\cos^2\theta\,d\psi)\Big)^2 + \\ c_2^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2 + c_3^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2\Big]$$

Doing the same for (s_1, s_3),

$$O(0): \frac{2m}{\Sigma}\Big[\frac{1}{H_2}\frac{1}{s_2^2}\Big(c_2\,dt - \frac{c_1c_2c_3}{c_2}(a\sin^2\theta\,d\phi + b\cos^2\theta\,d\psi)\Big)^2 + \\ c_1^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2 + c_3^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2\Big]$$

Doing the same for (s_1, s_2),

$$O(0): \frac{2m}{\Sigma}\Big[\frac{1}{H_3}\frac{1}{s_3^2}\Big(c_3\,dt - \frac{c_1c_2c_3}{c_3}(a\sin^2\theta\,d\phi + b\cos^2\theta\,d\psi)\Big)^2 + \\ c_1^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2 + c_2^2(b\sin^2\theta\,d\phi + a\cos^2\theta\,d\psi)^2\Big]$$

Example. (Four charge AdS_4 solution) We first present the explicit form of the solution and the gauge fields,

$$ds^2 = \frac{W}{\Sigma}\Big[-\frac{\Sigma(r^2+a^2\cos^2\theta-2mr)}{W^2}\Big(dt + \frac{2ma\sin^2\theta(rc_1c_2c_3c_4-(r-2m)s_1s_2s_3s_4)}{r^2+a^2\cos^2\theta-2mr}\,d\phi\Big)^2 \\ + \Sigma\Big(\frac{dr^2}{r^2+a^2-2mr} + d\theta^2 + \frac{r^2+a^2-2mr}{r^2+a^2\cos^2\theta-2mr}\sin^2\theta\,d\phi^2\Big)\Big]$$

Meanwhile, the gauge fields are given as

$$A_i = \frac{2m}{aW^2}\Big[(\frac{r_1r_2r_3r_4}{r_i} + ru^2)[c_is_ia\,dt - (a^2-u^2)(\frac{c_1c_2c_3c_4}{c_i}s_i - \frac{s_1s_2s_3s_4}{s_i}c_i)\,d\phi] \\ + 2mu^2[e_ia\,dt - (a^2-u^2)\frac{s_1s_2s_3s_4}{s_i}c_i\,d\phi]\Big]$$

where

$$e_i = \frac{c_1c_2c_3c_4s_1s_2s_3s_4}{c_is_i}(c_i^2+s_i^2) - c_is_i\Big(\sum_{\substack{j<k\\ j,k\neq i}} s_j^2s_k^2 + \frac{2s_1^2s_2^2s_3^2s_4^2}{s_i^2}\Big)$$

and

$$
\begin{aligned}
H_i =& 1+\frac{2mrs_i^2}{\Sigma} \quad c_i=\sqrt{1+s_i^2} \quad r_i = r+2ms_i^2 \\
W^2 =& (r_1r_2+u^2)(r_3r_4+u^2)-4mu^2(s_1s_2c_3c_4-c_1c_2s_3s_4)^2
\end{aligned}
$$

Preforming a Taylor expansion of the terms inside the warp factor with respect to (s_2, s_3, s_4) we find (up to $O(s_2^4, s_3^4, s_4^2)$ terms and subtracting the pure AdS terms)

$$
\begin{aligned}
O(0): &-\frac{\Sigma(r^2+a^2\cos^2\theta-2mr)}{(r_1r+u^2)(r^2+u^2)}\Big(dt+\frac{2mrc_1a\sin^2\theta}{r^2+a^2\cos^2\theta-2mr}\,d\phi\Big)^2 \\
&+\Sigma\Big(\frac{dr^2}{r^2+a^2-2mr}+d\theta^2+\frac{r^2+a^2-2mr}{r^2+a^2\cos^2\theta-2mr}\sin^2\theta\,d\phi^2\Big)
\end{aligned}
$$

Doing the same for (s_1, s_3, s_4),

$$
\begin{aligned}
O(0): &-\frac{\Sigma(r^2+a^2\cos^2\theta-2mr)}{(r_2r+u^2)(r^2+u^2)}\Big(dt+\frac{2mrc_2a\sin^2\theta}{r^2+a^2\cos^2\theta-2mr}\,d\phi\Big)^2 \\
&+\Sigma\Big(\frac{dr^2}{r^2+a^2-2mr}+d\theta^2+\frac{r^2+a^2-2mr}{r^2+a^2\cos^2\theta-2mr}\sin^2\theta\,d\phi^2\Big)
\end{aligned}
$$

Doing the same for (s_1, s_2, s_4),

$$
\begin{aligned}
O(0): &-\frac{\Sigma(r^2+a^2\cos^2\theta-2mr)}{(r_3r+u^2)(r^2+u^2)}\Big(dt+\frac{2mrc_3a\sin^2\theta}{r^2+a^2\cos^2\theta-2mr}\,d\phi\Big)^2 \\
&+\Sigma\Big(\frac{dr^2}{r^2+a^2-2mr}+d\theta^2+\frac{r^2+a^2-2mr}{r^2+a^2\cos^2\theta-2mr}\sin^2\theta\,d\phi^2\Big)
\end{aligned}
$$

Doing the same for (s_2, s_3, s_4),

$$
\begin{aligned}
O(0): &-\frac{\Sigma(r^2+a^2\cos^2\theta-2mr)}{(r_4r+u^2)(r^2+u^2)}\Big(dt+\frac{2mrc_4a\sin^2\theta}{r^2+a^2\cos^2\theta-2mr}\,d\phi\Big)^2 \\
&+\Sigma\Big(\frac{dr^2}{r^2+a^2-2mr}+d\theta^2+\frac{r^2+a^2-2mr}{r^2+a^2\cos^2\theta-2mr}\sin^2\theta\,d\phi^2\Big)
\end{aligned}
$$

Simplifying the first expression we found and subtracting the pure AdS part, we find

$$
\frac{2mr}{\Sigma H_1}(c_1\,dt-a\sin^2\theta\,d\phi)^2
$$

Ungauged Four Charge Ansatz

Using the knowledge from the coefficient determination process above and also educated guess work, I was able to arrive at the following form of the ungauged four charge ansatz.

$$
\begin{aligned}
ds^2 = \frac{W}{r^2 + a^2\cos^2\theta}\Big[&- dt^2 + (r^2 + a^2\cos^2\theta)\Big(\frac{dr^2}{r^2 + a^2 - 2mr} + d\theta^2\Big) \\
&+ (r^2 + a^2)\sin^2\theta \, d\phi^2 + \frac{W^2}{2m}\Big(\sum_{i=1}^{4} F_i A_i^2 + \sum_{i=1}^{4}(G_i - F_i)[A_i^{(\phi)}]^2\Big)\Big]
\end{aligned}
$$

where $A_i^{(\phi)}$ denotes the ϕ-part of the gauge field A_i and F_i and G_i each denote the following:

$$
\begin{cases}
F_i = \Big(\dfrac{r_1 r_2 r_3 r_4}{r_i} + r a^2\cos^2\theta + 2ma^2\cos^2\theta \, \dfrac{e_i}{c_i s_i}\Big)^{-1} s_i^4 \Big(\displaystyle\prod_{j\neq i}(s_i^2 - s_j^2)^{-1}\Big) \\[2ex]
G_i = \Big(\dfrac{r_1 r_2 r_3 r_4}{r_i} + r a^2\cos^2\theta + 2ma^2\cos^2\theta \, \dfrac{s_1 s_2 s_3 s_4 c_i^2}{c_1 c_2 c_3 c_4 s_i^2 - s_1 s_2 s_3 s_4 c_i^2}\Big)^{-1} s_i^4 \Big(\displaystyle\prod_{j\neq i}(s_i^2 - s_j^2)^{-1}\Big)
\end{cases}
$$

Gauged Four Charge Ansatz Construction

The ungauged four charge ansatz above compels us to start constructing the gauged four charge ansatz we've been building towards. Due to the elegance of the Wu form, the pure anti-de Sitter part has a trivial gauged generalisation, namely

$$
\begin{aligned}
ds^2 = \frac{W}{r^2 + a^2\cos^2\theta}\Big[&- \frac{(1 + g^2 r^2)(1 - a^2 g^2\cos^2\theta)}{1 - a^2 g^2} dt^2 + (r^2 + a^2\cos^2\theta)\Big(\frac{dr^2}{\Delta} + \frac{d\theta^2}{1 - a^2 g^2\cos^2\theta}\Big) \\
&+ \frac{r^2 + a^2}{1 - a^2 g^2}\sin^2\theta \, d\phi^2 + \frac{W^2}{2m}\Big(\sum_{i=1}^{4} F_i A_i^2 + \sum_{i=1}^{4}(G_i - F_i)[A_i^{(\phi)}]^2\Big)\Big]
\end{aligned}
$$

and the new gauge fields would be defined through the equality

$$
\begin{aligned}
\frac{W^2}{2m}A_i =& K_1\Big(\frac{r_1 r_2 r_3 r_4}{r_i} + 2ma^2\cos^2\theta\frac{e_i}{c_i s_i} + r a^2\cos^2\theta\Big) \\
&\times \Big[K_2 \, c_i s_i \, dt - K_3 \, a\sin^2\theta\Big(\frac{c_1 c_2 c_3 c_4 s_i}{c_i} - \frac{s_1 s_2 s_3 s_4 c_i}{s_i}\Big) d\phi\Big] \\
&+ K_4 \, 2ma^2\cos^2\theta\Big(\frac{c_1 c_2 c_3 c_4}{c_i^2}(e_i - c_i s_i) - \frac{s_1 s_2 s_3 s_4}{s_i^2}(e_i - c_i s_i)\Big) a\sin^2\theta \, d\phi
\end{aligned}
$$

Notice that we have a total of 5 undetermined elements, namely $(\Delta, K_1, K_2, K_3, K_4)$. We suggest the following methods for the determination process:

(i) Proper truncation limits for the two independent charge case and the pairwise identification case

(ii) Educated guesswork using symmetry and the original Wu paper

We will try to